

1 Week 9

Let m be a set function defined for all sets in a σ -algebra $\mathcal{A}$ with values in $[0, \infty]$. Assume m is countably additive over countable disjoint collections of sets in $\mathcal{A}$.

Question 1.1 (Royden 2.1). Prove that if A and B are two sets in $\mathcal{A}$ with $A \subseteq B$, then $m(A) \leq m(B)$. This property is called *monotonicity*.

Proof. Since $A = A \sqcup (B \setminus A)$ we have $m(B) = m(A) + m(B \setminus A)$. From this equality, it is evident that A may not have infinite measure unless B does as well, so the inequality $m(A) \leq m(B)$ is obvious for when B has infinite measure. For B with finite measure, we have $m(A) \leq m(A) + m(B \setminus A) = m(B)$ since $m(B \setminus A) \geq 0$. $\square$

Question 1.2 (Royden 2.2). Prove that if there is a set A in the collection $\mathcal{A}$ for which $m(A) < \infty$ then $m(\emptyset) = 0$.

Proof. Since $\emptyset \subseteq A$, we have by Royden 2.1 that $m(\emptyset) \leq m(A)$ is finite. Since $\emptyset = \emptyset \sqcup \emptyset$, we have $m(\emptyset) = m(\emptyset) + m(\emptyset) \Rightarrow m(\emptyset) = 0$. $\square$

Question 1.3 (Royden 2.3). Let $\{E_k\}_{k=1}^{\infty}$ be a countable collection of sets in $\mathcal{A}$. Prove that $m(\bigcup_{k=1}^{\infty} E_k) \leq \sum_{k=1}^{\infty} m(E_k)$.

Proof. Define the collection of sets $\{F_k\}_{k=1}^{\infty}$ by $F_k = E_k \setminus \bigcup_{i=1}^{k-1} E_i \in \mathcal{A}$. It is clear that $\bigcup_{k=1}^{\infty} F_k \subseteq \bigcup_{k=1}^{\infty} E_k$. For the reverse inclusion, given an element $x \in \bigcup_{k=1}^{\infty} E_k$, we take the smallest natural n for which $x \in E_n$ and note that $x \in F_n$ by construction of the F_k 's. Therefore the two unions are equal. Furthermore, given positive integers $n < m$, we have that $F_n \subseteq E_n$ and $F_m \cap E_n = \emptyset$ by construction, so $F_n \cap F_m = \emptyset$. Thus the collection $\{F_k\}_{k=1}^{\infty}$ is a disjoint collection of sets in $\mathcal{A}$ and we have

$$m\left(\bigcup_{k=1}^{\infty} E_k\right) = m\left(\bigcup_{k=1}^{\infty} F_k\right) = \sum_{k=1}^{\infty} m(F_k) \leq \sum_{k=1}^{\infty} m(E_k)$$

as required. $\square$

Question 1.4 (Royden 2.4). A set function c , defined on all subsets of $\mathbb{R}$ is defined as follows. Define $c(E)$ to be ∞ if E has infinitely many members and $c(E)$ to be the number of elements in E if E is finite; define $c(\emptyset) = 0$. Show that c is a countably additive and translation invariant set function. This set function is called the **counting measure**.

Proof. $\square$

Question 1.5 (Royden 2.8). Let B be the set of rational numbers in the interval $[0, 1]$, and let $\{I_k\}_{k=1}^{\infty}$ be a finite collection of open intervals that covers B . Prove that $\sum_{k=1}^n m^*(I_k) > 1$.

Proof. $\square$

Question 1.6 (Royden 2.10). Let A and B be bounded sets for which there is an $\alpha > 0$ such that $|a - b| > \alpha$ for all $a \in A, b \in B$. Prove that $m^*(A \cup B) = m^*(A) + m^*(B)$.

Proof. $\square$

Question 1.7 (Royden 2.11). Prove that if a σ -algebra of subsets of $\mathbb{R}$ contains intervals of the form (a, ∞) , then it contains all intervals.

Proof. $\square$

Question 1.8 (Royden 2.19). Let E have finite outer measure. Show that if E is not measurable, then there is an open set $\mathcal{O}$ containing E that has finite outer measure and for which

$$m^*(\mathcal{O} \setminus E) > m^*(\mathcal{O}) - m^*(E).$$

Proof.

□

Question 1.9 (Royden 2.20: Lebesgue). Let E have finite outer measure. Show that E is measurable if and only if for each open, bounded interval (a, b) ,

$$b - a = m^*((a, b) \cap E) + m^*((a, b) \setminus E).$$

Proof.

□

Question 1.10 (Royden 2.24). Show that if E_1 and E_2 are measurable, then

$$m(E_1 \cup E_2) + m(E_1 \cap E_2) = m(E_1) + m(E_2).$$

Proof.

□

Question 1.11 (Royden 2.25). Show that the assumption that $m(B_1) < \infty$ is necessary in part (ii) of the theorem regarding continuity of measure.

Proof.

□

Question 1.12 (Royden 2.27). Let $\mathcal{M}'$ be any σ -algebra of subsets of $\mathbb{R}$ and m' a set function on $\mathcal{M}'$ which takes values in $[0, \infty]$, is countably additive, and such that $m'(\emptyset) = 0$.

- (i) Show that m' is finitely additive, monotone, countably monotone, and possesses the excision property.
- (ii) Show that m' possesses the same continuity properties as Lebesgue measure.

Proof.

□

Question 1.13 (Royden 2.28). Show that continuity of measure together with finite additivity of measure implies countable additivity of measure.

Proof.

□